COURSE INFORMATION SHEET

# Basics of Electrical Engineering

**Course Code: EE24101**
**Course Title: Basics of Electrical Engineering**
**Pre-requisite(s):**
**Co- requisite(s): Basic Sciences**
**Credits:** L: 2 T: 1 P:
**Class schedule per week: 03**
**Class: B.Tech.**
**Semester / Level: I/01**
**Branch: All**

## COURSE OBJECTIVES

This course envisions to impart to students to:

| | |
|---|---|
| **1.** | realize the electrical signals, elements, and their properties. |
| **2.** | understand the mathematical representation of AC, DC signals and theorems/laws for solving electrical circuits with variations of voltage and frequency. |
| **3.** | perceive the 3-phase AC signal representation and 3-phase circuit analysis for balanced and unbalanced condition. |
| **4.** | understand the characteristics of magnetic material and analysis of magnetic circuits. |

## COURSE OUTCOMES (COs)

After the completion of this course, students will be able to:

| | |
|---|---|
| **CO1** | explain the voltage, current signals and their characteristics in electrical circuit elements. |
| **CO2** | apply the theorems/laws for electrical circuit analysis. |
| **CO3** | solve the electrical circuits for variable voltage and frequency to observe the resonance, power and power factor in the electric circuit. |
| **CO4** | analyze the 1-phase and 3-phase AC balanced and unbalanced circuits |
| **CO5** | apply the concept of magnetic circuits for magnetic circuit analysis. |

**SYLLABUS**

| MODULE | (NO. OF LECTURE HOURS) |
|---|---|
| **Module – I**<br>**Introduction:** Importance of Electrical Engineering in day-to-day life, Electrical elements, properties (linear, non-linear, unilateral, bilateral, lumped and distributed, etc.) and their classification, Ideal and Real Sources, Source Conversion, Star-Delta conversion, KCL and KVL, Mesh current and Nodal voltage method. | **8** |
| **Module – II**<br>**D.C. Circuits**: Steady state analysis with independent and dependent sources; Series and Parallel circuits.<br>**Circuit Theorems:** Superposition, Thevenin's, Norton's, and Maximum Power Transfer theorems for Independent and Dependent Sources applied to DC circuits. | **8** |
| **Module – III**<br>**Single-phase AC Circuits:** Common signals and their waveforms, RMS and Average value. Form factor & Peak factor of a sinusoidal waveform. **Series Circuits:** Impedance of Series circuits. Phasor diagram. Active Power. Power factor. Power triangle. **Parallel Circuits:** Admittance method, Phasor diagram, Power and Power factor Power triangle, Series-parallel Circuit, Power factor improvement, Circuit Theorems applied to AC circuits.<br>**Series and Parallel Resonance**: Resonance curve, Q–factor, Dynamic Impedance, and Bandwidth. | **12** |
| **Module – IV**<br>**Three-Phase AC Circuits:** Importance and use of a 3-phase network, types of 3-phase connections- Star and Delta, Line and Phase relations for Star and Delta connection, Phasor diagrams, Power relations, analysis of balanced and unbalanced 3-phase circuits, Measurement of Power in 3-phase star and delta network. | **6** |
| **Module – V**<br>**Magnetic Circuits:** Introduction, Series-parallel magnetic circuits, Analysis of Linear and Non-linear magnetic circuits, Energy storage, A.C. excitation, Eddy currents and Hysteresis losses.<br>**Coupled Circuits:** Dot rule, Self and mutual inductances, Coefficient of coupling, working of transformer. | **6** |

**TEXTBOOKS:**

1. W. H. Hayt, Jr J. E. Kemmerly and S. M. Durbin, Engineering Circuit Analysis, $7^{th}$ Edition TMH, 2010.
2. Hughes, Electrical Technology, Revised by McKenzie Smith, Pearson.
3. Fitzgerald and Higginbotham, Basic Electrical Engineering, McGraw Hill Inc, 1981

**REFERENCE BOOKS:**

1. D. P. Kothari and I. J. Nagrath, Basic Electrical Engineering, 3rd Edition, TMH, New Delhi, 2009.
2. Electrical Engineering Fundamental, Vincent Del Toro, Prentice Hall, New Delhi.
3. Rajendra Prasad, Fundamentals of Electrical Engineering, $2^{nd}$ Edition, PHI, New Delhi, 2011.
4. Raymond A. DeCarlo, Prn-Min Lin, Linear Circuit Analysis Time Domain, Phasor and Laplace Transform Approaches, $2^{nd}$ Edition, Oxford University, 2001
5. Abhijit Chakrabarti, Sudipta Nath, Chandan Kumar Chanda, Basic Electrical Engineering, Tata McGraw Hill Publication, 2009.

**GAPS IN THE SYLLABUS (TO MEET INDUSTRY/PROFESSION REQUIREMENTS)**

1. Application of principles of magnetic circuits to electrical machines like transformers, generators and motors.
2. Field applications of three phase equipment and circuits in power system.
3. Applications of circuit theorems in electrical and electronics engineering

**POS MET THROUGH GAPS IN THE SYLLABUS: 3, 4, 12**

**TOPICS BEYOND SYLLABUS/ADVANCED TOPICS/DESIGN**

1. Concepts of electric, magnetic and electromagnetic fields.
2. 3 - Φ power generation, transmission, and distribution.
3. Power factor improvement for three phase systems.
4. Utility of reactive power for creation of electric and magnetic fields.

**POS MET THROUGH TOPICS BEYOND SYLLABUS/ADVANCED TOPICS/DESIGN:** 2, 3, 4, 12

**COURSE OUTCOME (CO) ATTAINMENT ASSESSMENT TOOLS & EVALUATION PROCEDURE**

**DIRECT ASSESSMENT**

| Assessment Tool | % Contribution during CO Assessment |
|---|---|
| Quiz (s) | 10 |
| End Semester Examination | 50 |
| Mid Semester Examination | 25 |
| Assignment | 10 |
| Teacher Assessment | 05 |

| Continuous Internal Assessment | % Distribution |
|---|---|
| Assignment | 10 |

| | |
|---|---|
| Teacher Assessment | 05 |

| Assessment Components | CO1 | CO2 | CO3 | CO4 | CO5 |
|---|---|---|---|---|---|
| Continuous Internal Assessment | | | | | |
| Semester End Examination | | | | | |

**INDIRECT ASSESSMENT**

**1. Student Feedback on Course Outcome**

**COURSE DELIVERY METHODS**

| | |
|---|---|
| **CD1** | Lecture by use of boards/LCD projectors/OHP projectors |
| **CD2** | Tutorials/Assignments |
| **CD3** | Seminars |
| **CD4** | Mini projects/Projects |
| **CD5** | Laboratory experiments/teaching aids |
| **CD6** | Industrial/guest lectures |
| **CD7** | Self- learning such as use of NPTEL materials and internets |
| **CD8** | Simulation |

**MAPPING BETWEEN COURSE OUTCOMES AND POs and PSOs**

| | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 | PSO3 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 2 | 1 | | 1 | 1 | | | | 1 | | | 2 | 1 | |
| CO2 | 3 | 3 | 3 | 1 | 2 | 1 | | | | 1 | | | 3 | 3 | |
| CO3 | 3 | 3 | 3 | 2 | 2 | 1 | 1 | | | 1 | | 1 | 3 | 3 | |
| CO4 | 3 | 3 | 3 | 2 | 2 | 1 | | | | 1 | | | 3 | 3 | |
| CO5 | 3 | 3 | 3 | 2 | 2 | 1 | | | | 1 | | | 3 | 3 | |

**Grading: No correlation – 0, Low correlation - 1, Moderate correlation – 2, High Correlation - 3**

**MAPPING BETWEEN COURSE OUTCOMES AND COURSE DELIVERY METHOD**

| Course Outcomes | Course Delivery Method |
|---|---|
| **CO1** | CD1, CD2, CD5 |
| **CO2** | CD1, CD2, CD4, CD5, CD7 |
| **CO3** | CD1, CD2, CD5, CD7, CD8 |
| **CO4** | CD1, CD2, CD5, CD7, CD8 |
| **CO5** | CD1, CD2, CD4, CD5, CD7, CD8 |